

Air Quality Analysis and Prediction using Machine Learning

A data-driven approach to understanding and forecasting PM2.5 pollution levels

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Introduction

What is Air Pollution?

Air pollution is the presence of harmful substances — gases, particles, and biological matter — in the atmosphere. Rapid urbanisation and industrialisation have made it a critical global health concern.

Why PM2.5 Matters

PM2.5 refers to fine particulate matter smaller than 2.5 micrometres. These tiny particles penetrate deep into the lungs, causing respiratory disease, heart conditions, and long-term organ damage.

 7 Million

Deaths per year linked to air pollution (WHO)

 Prediction

ML models can forecast pollution 24–48 hrs ahead, enabling early warnings

 Data-Driven

Weather variables like temperature and wind speed are strong predictors of PM2.5

Literature Survey

Key research papers that informed this project:

1

Comprehensive comparison of correction techniques for low-cost air quality sensors: the impact of device type and deployment environment-Ildris Hayward, Nicholas A. Martin², Valerio Ferracci, Mohsen Kazemimanesh , Simon Jude , Christopher Walton , Zaheer Ahmad Nasir & Prashant Kumar,

2

Application of complementary air quality exposure assessment methods in a complex industrial-urban environment: a case study from Venafro Valley, Italy- Cristina Mangia¹, Elisa Bustaffa, Marco Cervino, Fabrizio Bianchi ,Liliana Cori , Fabrizio Minichilli

3

Air Pollution in Small Urban Settlements Cannot be Ignored: A Case Study of Jimunai, Northwest China- Xiaoling Zhong, Pengfei Chen, Shichang Kang, Huijun Zheng, Junming Guo, Zhaofu Hu, Xiaobo Wu, Chaman Gul, Umesh Chandra Dumka

4

Air Quality Assessment and Source Speciation of PM_{2.5} in Florida's Everglades Agricultural Area- Gamal El Afandi, Olga Bolden-Tiller, Desmond Mortley, Lucy Asare-Baah, Randall Miller, Salem Ibrahim, Santosh Sapkota, Abhinav Yadav, Sushrisangita Sahoo, Vijaya Rangari

5

Air quality prediction using multisource remote sensing data integration with hybrid deep learning framework-S. Kalaiselvi^{1,2}, V. Anitha, V. Manimaran & Thomas Samraj Lawrence

Dataset Overview



Source & Structure

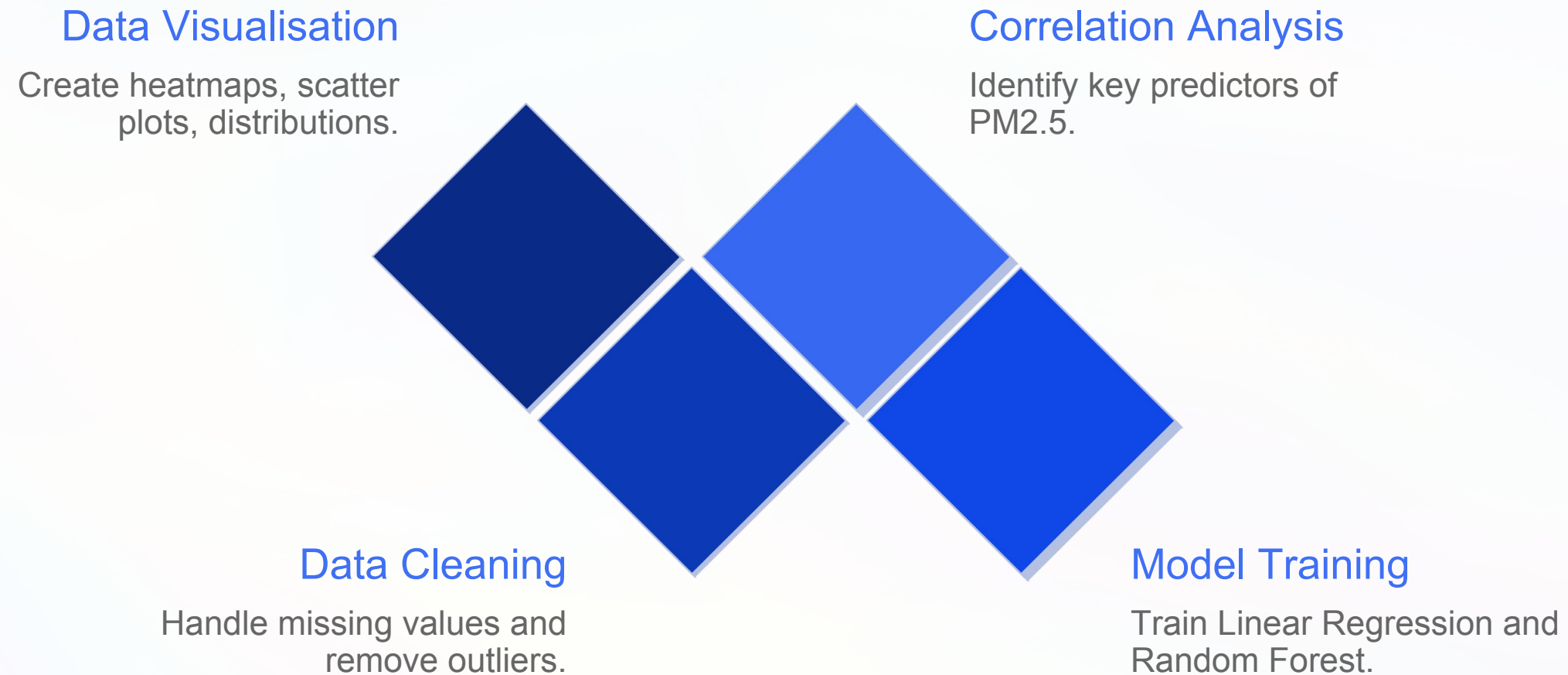
The dataset contains hourly environmental readings collected from urban monitoring stations. It includes both meteorological features and the target pollution metric.

Feature	Type	Role
Temperature	Numeric	Predictor
Humidity	Numeric	Predictor
Pressure	Numeric	Predictor
Wind Speed	Numeric	Predictor
PM2.5	Numeric	Target

i Missing values were handled using median imputation. Outliers were capped at the 95th percentile.

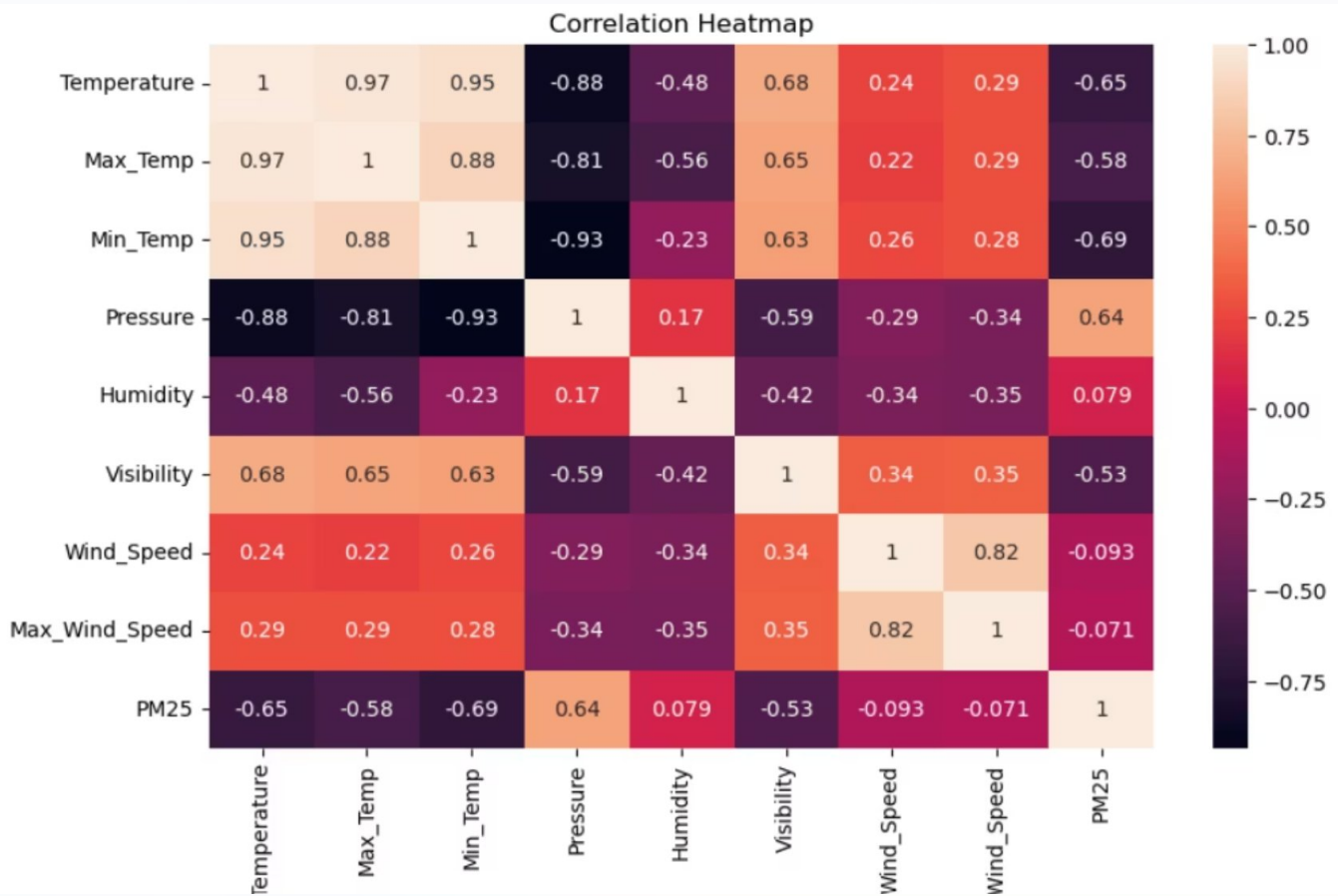
Methodology

A structured pipeline was followed from raw data to trained prediction models.



Each stage ensured data quality and model reliability, producing interpretable and accurate predictions of PM2.5 concentration.

Data Visualisation



Reading the Heatmap

A correlation heatmap shows how strongly each weather feature is related to PM2.5. Values close to **+1** indicate a positive relationship, while values near **-1** indicate an inverse relationship.

Wind Speed ↔ PM2.5

Strong negative correlation — higher wind disperses pollutants

Humidity ↔ PM2.5

Positive correlation — moist air traps particulates near ground level

Temperature ↔ PM2.5

Moderate negative correlation — warmer air promotes dispersion



Key Insights from the Data



Wind Cleans the Air

Higher wind speed strongly reduces PM2.5 by dispersing fine particles across a wider area.



Humidity Traps Pollution

High humidity causes PM2.5 to linger near the surface, increasing measured concentrations.



Temperature: Moderate Effect

Warmer temperatures aid vertical mixing of air, offering a modest reduction in ground-level PM2.5.



Pressure: Weak Signal

Atmospheric pressure showed low correlation with PM2.5 and contributed less to model accuracy.

Machine Learning Models

Linear Regression

A foundational statistical model that finds the best-fit straight line between input features and PM2.5. Simple to interpret, but assumes a linear relationship — which real-world pollution data often violates.



Struggles with non-linear patterns and outliers common in environmental data.

Random Forest

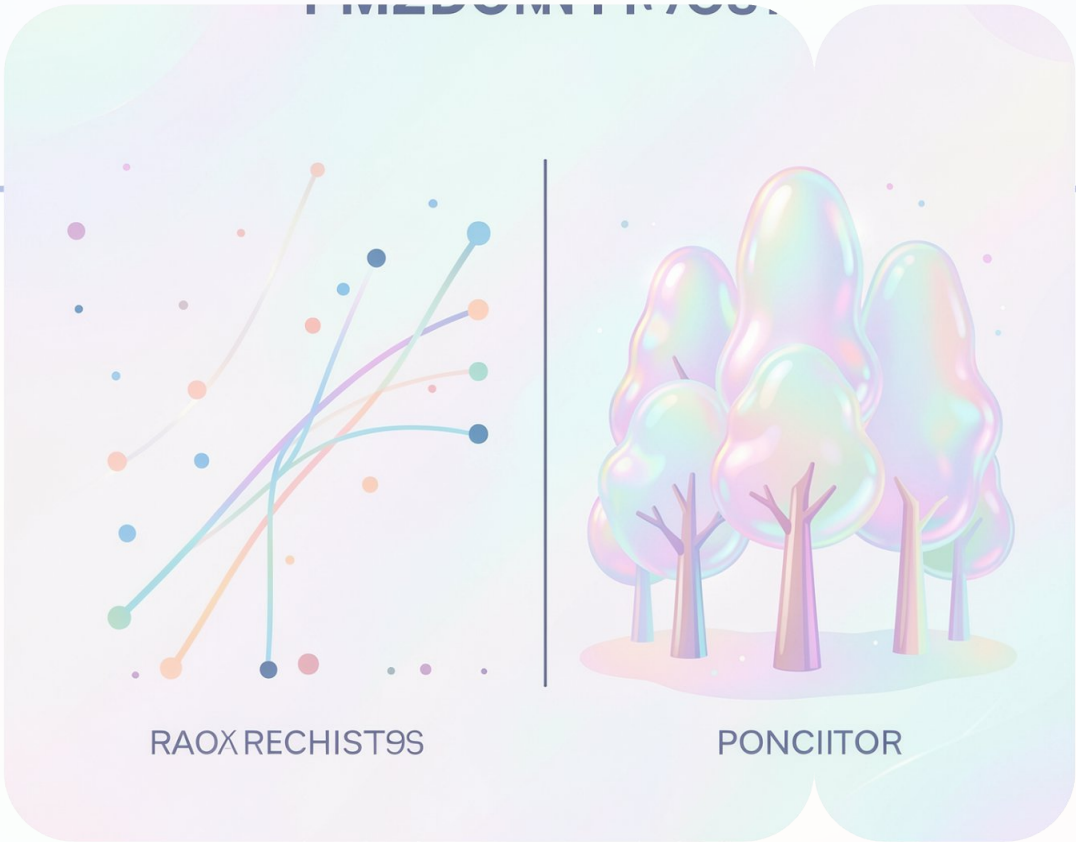
An ensemble method that builds hundreds of decision trees and averages their predictions. It naturally handles non-linearity, missing values, and complex interactions between weather variables.



Captures complex patterns and is robust to noise — ideal for PM2.5 prediction.

Linear Regression

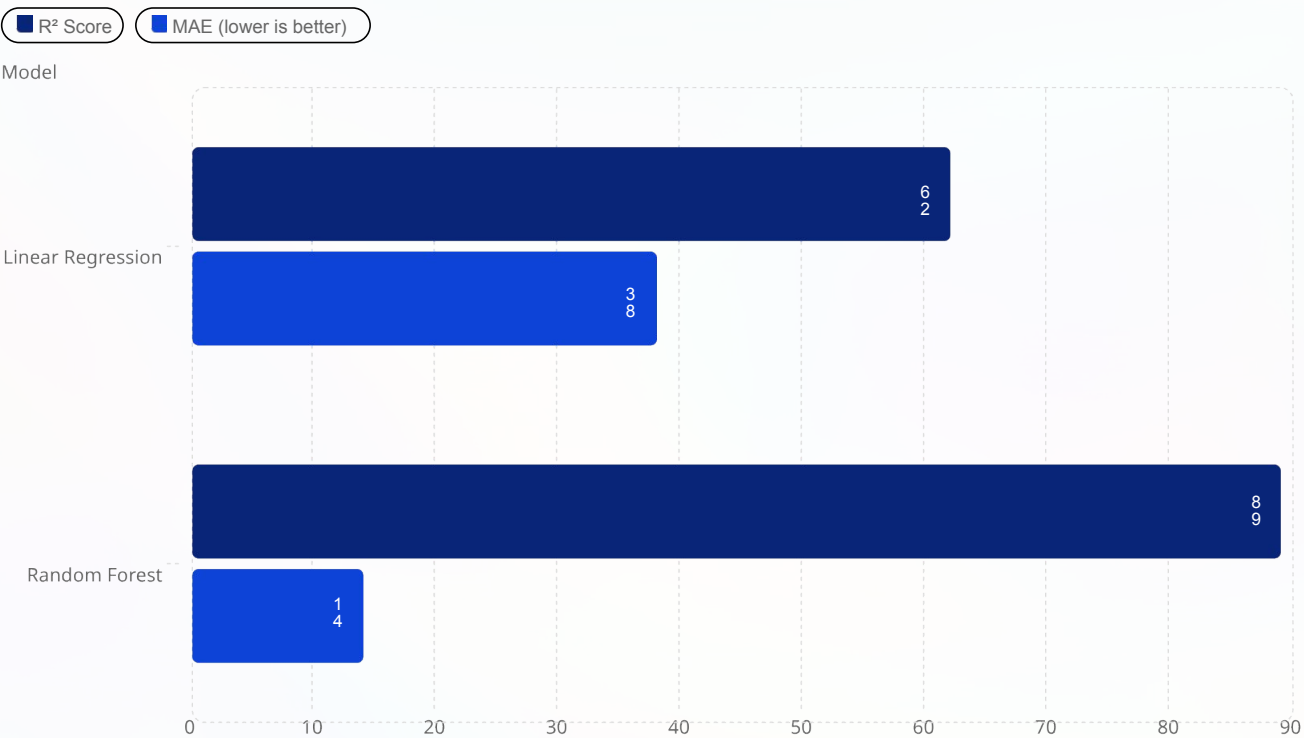
One line, simple, lower accuracy



Random Forest

Many trees combined, complex, higher accuracy

Model Comparison



Why Random Forest Wins

Random Forest achieved an **R² score of ~0.89** versus Linear Regression's ~0.62 — a significant improvement in predictive accuracy.

- Handles **non-linear** relationships between weather and pollution
- Less sensitive to **outliers** in sensor readings
- Combines **multiple trees** to reduce prediction error

Conclusion & References

What We Learned

Weather Matters

Meteorological variables are reliable predictors of PM2.5 levels

ML Helps

Machine learning can forecast pollution with high accuracy

RF Wins

Random Forest outperforms Linear Regression for this task